

Inverse-Designed Suspended Silicon-Nitride Photonic-Crystal Cavities for High-Efficiency Atom–Photon Transduction at the Rubidium D2 Line

First Author,^{1,*} Second Author,¹ and Senior Author¹

¹*Department of Physics, Institution, City, Country*

(Dated: June 15, 2026)

Scaling neutral-atom quantum processors into modular networks requires deterministic, high-efficiency interfaces that map a trapped atom’s state onto a propagating photon. We present a suspended silicon-nitride one-dimensional photonic-crystal nanobeam cavity, inverse-designed to be resonant with the ^{87}Rb D2 line (384.231 THz, 780 nm), that directly maximizes the atom-to-waveguide transduction efficiency rather than the bare quality factor. Using differentiable adjoint finite-difference time-domain simulation, we co-optimize per-hole lattice constants, hole sizes, and free-form closed spline hole shapes, seeded by a Bloch band-engineered mirror cell; a single conformal in-plane scale then tunes the resonance onto the atomic line while preserving the dimensionless photonic design. From the classical electromagnetic fields we extract an interface efficiency $\eta = \beta C / (C+1) \eta_{\text{fb}} = 0.84$, with waveguide branching ratio $\beta = 0.89$, cooperativity $C = 24$, single-photon coupling $g/2\pi = 1.09$ GHz at a realistic 250 nm trap height, and mode volume $0.87 (\lambda/n)^3$ — a tenfold gain in η over a hand-tuned baseline. The design is robust across a fabrication corner grid. These figures of merit are electromagnetically derived scalar proxies, not the output of a full cavity-QED simulation.

I. INTRODUCTION

Arrays of individually trapped neutral atoms have emerged as a leading platform for quantum computation and simulation. Optical tweezers assemble defect-free registers of hundreds of identical qubits [1, 2], whose states are stored in long-lived hyperfine or clock transitions and entangled through strong, switchable Rydberg interactions with gate fidelities now exceeding 99% [3]. Coherent transport of trapped atoms enables reconfigurable, nonlocal connectivity within a single processor [4]. Relative to trapped-ion processors—where a shared motional bus couples qubits but ultimately limits the size of a single crystal [5]—neutral-atom arrays offer a more favorable route to large qubit numbers and dynamically rewirable connectivity, at the cost of weaker native long-range coupling between distant registers.

Bridging that gap, and scaling beyond a single processor, calls for a *photonic* interconnect: a quantum channel that converts a stationary atomic qubit into a flying optical photon and back [6]. Cavity quantum electrodynamics provides the enabling mechanism. An optical resonator enhances the atom’s emission into a well-defined mode and routes it into a waveguide, so that two remote nodes can be heralded into an entangled state by interfering their emitted photons [7, 8]. Integrated nanophotonic implementations are especially attractive for atom arrays because they are compact, support many parallel channels on one chip, and can be aligned to the atomic geometry [9, 10].

The rate and fidelity of such an interconnect are governed by a single figure of merit: the efficiency η with which an intracavity excitation is delivered into the col-

lection fiber. Two-node entanglement rates scale as η^2 , and the residual infidelity of cavity-mediated protocols falls only as $\ln C/C$ with the cooperativity C [7, 11]. Reaching high η simultaneously demands (i) a large Purcell enhancement, so the atom preferentially emits into the cavity; (ii) strong *over-coupling*, so the cavity decays predominantly into the collection waveguide rather than into scattering or absorption; and (iii) precise resonance with the atomic transition. Pioneering single-atom nanophotonic experiments reached cooperativities of order unity but extracted only a small fraction of the emission into a usable mode—for example $\eta \sim 0.07$ in the first photonic-crystal-waveguide cavity coupled to a single ^{87}Rb atom [12]—and subsequent integrated nodes identified the waveguide out-coupling fraction as a dominant loss [10, 11, 13, 14]. Closing this gap is the key photonic bottleneck to scalable atom-array interconnects.

We target a specific architecture for that interconnect, sketched in Fig. 1(a): an array of suspended photonic-crystal “fingers” cantilevered beyond an etched substrate edge. Atoms held in optical tweezers are transported over the series of cavities, and at each finger the atomic state is transduced into a flying photonic qubit that is collected into fiber. Realizing this picture at scale requires a cavity that is simultaneously low-mode-volume, strongly over-coupled, fabrication-robust, and *exactly* resonant with the atomic line. Silicon nitride is a natural host: it is transparent and low-loss at 780 nm, and a released nanobeam exposes the cavity field to the vacuum mode where the atom is trapped. The design space—hole spacing, hole size, and hole shape, varied independently along the beam—is high-dimensional and nonintuitive, which makes it well suited to adjoint-based inverse design rather than hand tuning [15, 16], building on the deterministic gentle-confinement principles developed for ultra-high- Q nanobeams [17–19].

In this work we design such a cavity. Starting from a

* first.author@institution.edu

Bloch band-engineered seed whose mirror gap is centered on the ^{87}Rb D2 line, we use differentiable finite-difference time-domain (FDTD) simulation to co-optimize the per-hole lattice constant, the hole size, and a free-form closed spline hole shape, with a loss function built directly on the transduction efficiency η rather than on Q or C alone. A final conformal in-plane scale tunes the resonance onto the atomic line. The resulting device reaches $\beta = 0.89$ and $\eta = 0.84$ —a tenfold improvement in η over a hand-tuned baseline and an order of magnitude above the efficiency of single-atom nanocavity demonstrations—while remaining single-mode and robust across a fabrication corner grid. We emphasize at the outset that all atom-photon figures of merit reported here are scalar proxies extracted from classical electromagnetic fields; we make no claim about gate, readout, or entanglement fidelities, which would require a full cavity-QED treatment beyond the present scope.

II. METHODS

A. Physical platform

The device is a suspended SiN nanobeam of thickness 150 nm and width 500 nm; both are fixed fabrication parameters set by the deposited film and the lithography, and the thickness places the first guided band edge so that the mirror gap can bracket 384 THz. A one-dimensional array of etched holes forms a photonic-crystal mirror on either side of a central defect. The cavity is deliberately asymmetric: a short coupling mirror on the output ($-x$) side and a long reflector on the back ($+x$) side, so that the intracavity field leaks preferentially into a single output waveguide. The atom is trapped in the retro-reflected tweezer antinode 200–300 nm above the top surface, above the Casimir–Polder–limited approach distance; we evaluate all coupling quantities at the nominal 250 nm height.

B. Figures of merit

All atom-photon quantities below are computed as scalar proxies from the simulated classical fields; they are not outputs of a master-equation or quantum-trajectory calculation. The single-photon coupling rate between an atom of dipole moment μ and the cavity vacuum field is

$$g = \mu \sqrt{\frac{\omega_c}{2\hbar\epsilon_0 V}}, \quad (1)$$

with V the mode volume. Equivalently, sampling the simulated mode at the atom position $\mathbf{r}$,

$$g(\mathbf{r}) = \mu \sqrt{\frac{\omega_c u(\mathbf{r})}{2\hbar\epsilon_0 \int \epsilon |\mathbf{E}|^2 dV}}, \quad u(\mathbf{r}) = |E_y(\mathbf{r})|^2, \quad (2)$$

which we use directly so that g reflects the true evanescent field at the trap height; $\mu = 3.58 \times 10^{-29}$ C m for the ^{87}Rb D2 transition. The mode volume and Purcell factor are

$$V = \frac{\int \epsilon |\mathbf{E}|^2 dV}{\max(\epsilon |\mathbf{E}|^2)}, \quad F_P = \frac{3}{4\pi^2} \frac{Q}{V}, \quad (3)$$

with V expressed in $(\lambda/n)^3$. The total cavity energy-decay rate is $\kappa = \omega_c/Q$, and the cooperativity is

$$C = \frac{4g^2}{\kappa\gamma}, \quad (4)$$

where $\gamma = 2\pi \times 6.07$ MHz is the ^{87}Rb D2 natural linewidth. In the bad-cavity (Purcell) regime $\kappa \gg g \gg \gamma$ [20], an excited atom emits into the cavity mode rather than into free space with probability $C/(C+1)$. The fraction of cavity decay that exits through the intended collection waveguide is the branching ratio

$$\beta = \frac{\kappa_{\text{wg}}}{\kappa_{\text{tot}}}, \quad (5)$$

obtained from the directional radiated flux. Combining these, the efficiency with which the atomic excitation is delivered as a collected photon—the figure of merit we optimize—is

$$\boxed{\eta = \beta \frac{C}{C+1} \eta_{\text{fib}}}, \quad (6)$$

with $\eta_{\text{fib}} = 0.99$ a fixed waveguide-to-fiber collection factor. Three distinct levers thus set η : the Purcell factor $C/(C+1)$ (emit into the cavity), the over-coupling β (route into the waveguide), and the collection η_{fib} . Crucially, because $C/(C+1)$ saturates once $C \gtrsim 10$, the relevant optimum maximizes η , *not* Q or C in isolation: pushing Q ever higher in a closed cavity drives $\beta \rightarrow 0$ and $\eta \rightarrow 0$ [21]. The residual infidelity of cavity-mediated entanglement scales as $\ln C/C$, so $C \sim 20$ already places the design in the regime of leading demonstrations. Finally, the photon is only emitted into the cavity if the resonance sits within roughly a linewidth of the atomic transition, $|f_{\text{res}} - f_{\text{D2}}| \lesssim \kappa/2\pi$; meeting this condition motivates the final trim.

C. Inverse design

We optimize the geometry with differentiable FDTD using the native adjoint capability of TIDY3D [15, 22]: a single forward frequency-domain simulation and one adjoint simulation yield the gradient of the objective with respect to every geometric degree of freedom. The cavity is parameterized by (i) a per-hole lattice constant a_i , so the hole placement schedule is free; (ii) per-hole in-plane scales $(s_{x,i}, s_{y,i})$; and (iii) a closed, free-form hole shape

described by a periodic cubic B-spline of the radius as a function of polar angle. The radial (star-shaped) parameterization is closed and provably non-self-intersecting for any positive control radii, and x/y mirror symmetry reduces the twelve control points to four free radii per shape; at unit radius the hole reduces to an ellipse, so the seed is reproduced exactly. A fixed five-anchor blend (taper, mirror, and defect anchors) ties the per-hole shapes into a smooth schedule along the beam.

The loss combines the physics objective with fabrication-aware geometric barriers,

$$\mathcal{L} = -\log \tilde{\eta} + w_f \left(\frac{\hat{f} - f_{\text{D2}}}{\Delta f} \right)^2 + \mathcal{P}_{\text{geom}}, \quad (7)$$

where $\tilde{\eta}$ is a differentiable surrogate for Eq. (6) built from a softmax-peak-picked waveguide-to-total power ratio (for β) and a calibrated cooperativity proxy, $\hat{f}$ is the soft-resonance frequency, and $\mathcal{P}_{\text{geom}}$ collects fabrication-aware barriers enforcing a minimum dielectric gap, side-rail width, spline curvature radius, and lattice smoothness (a gentle-confinement prior). Only frequency-domain field and mode monitors enter $\mathcal{L}$, since these are the differentiable quantities. We minimize $\mathcal{L}$ with Adam in two stages—first the lattice and hole-size layout, then the spline shapes—seeded by a Bloch band-structure design in which the mirror unit cell is sized so that its band gap is centered on the D2 line. Every candidate is independently verified outside the gradient loop by a time-domain FDTD simulation that extracts f_{res} and Q from the ring-down, the directional quality factors and hence β , the mode volume, and the field-sampled g , C , and η of Eqs. (2) and (4)–(6).

D. Resonance trim

A converged design generically lands tens of GHz from the atomic line. Because the in-plane geometry can be rescaled but the film thickness cannot, we tune the resonance with a single global conformal in-plane scale s applied to all hole positions, lattice constants, hole sizes, and spline vertices. Frequency scales approximately as $1/s$, but only approximately—the fixed thickness and the dispersive SiN index break exact scale invariance—so we measure $f_{\text{res}}(s)$ with a two-point bracket, interpolate linearly to the target, and apply one Newton refinement. Because a conformal scale preserves the *dimensionless* photonic design (the mirror gap and the cavity mode shift together), the trim changes β , C , and η by less than 1%.

E. Simulation parameters

Forward and verification simulations use the measured dispersive SiN index, a production mesh of $0.01 \mu\text{m}$ (about 15 steps per wavelength), an 8 ps run time (one to two cavity ring-downs), perfectly matched layers on

all boundaries, and the y/z mirror symmetries of the E_y mode. We note here a numerical caveat that bears on the absolute resonance placement: while β , η , and the mode profile are mesh-robust, the *absolute* resonant frequency varies by $\sim 80 \text{ GHz}$ as the mesh is refined from 0.012 to $0.006 \mu\text{m}$, exceeding the trim tolerance. We therefore fix one production mesh for the trim and for all designs compared below, and note that the residual frequency uncertainty is absorbed in practice by the same $\sim 0.1\%$ in-plane scale knob used for the trim.

III. RESULTS

A. Final device

Figure 1 shows the optimized cavity and its intended setting. The $25 \mu\text{m}$ -long nanobeam [Fig. 1(b)] comprises a short output coupling mirror, a graded defect that smoothly localizes the mode [Fig. 1(c)], and a long back reflector; the optimizer has shaped both the lattice schedule and the individual hole profiles. The released geometry exposes the mode to the trap region 250 nm above the surface [Fig. 1(d)].

B. Inverse-design trajectory

Figure 2 traces the optimization. The objective [Eq. (7)] falls from its initial value as the optimizer first opens the output port—rapidly raising β and hence η —before the frequency penalty pulls the resonance back toward the atomic line and the shape stage refines the mode; the intermediate excursion in $\mathcal{L}$ is precisely this frequency penalty acting when the resonance wanders, an honest signature of the multi-objective landscape. The structure snapshots [Fig. 2(b)–(d)] show the hole schedule evolving from the band-engineered seed to the final graded profile.

C. Mode structure and band engineering

The verified mode [Fig. 3(a)–(c)] is a single, tightly localized state centered on the defect, with an evanescent tail that reaches the 250 nm trap height. Its photonic origin is shown in Fig. 3(d): the mirror unit cell exhibits a transverse-electric photonic band gap [23] that brackets the D2 frequency, providing in-plane confinement, while the graded defect raises a band edge to the D2 line to localize the mode. This is the band-engineering that the seed supplies and that the inverse design preserves.

D. Performance and design lineage

The final trimmed device achieves a verified resonance of $f_{\text{res}} = 384.231 \text{ THz}$, on the ^{87}Rb D2 line to within

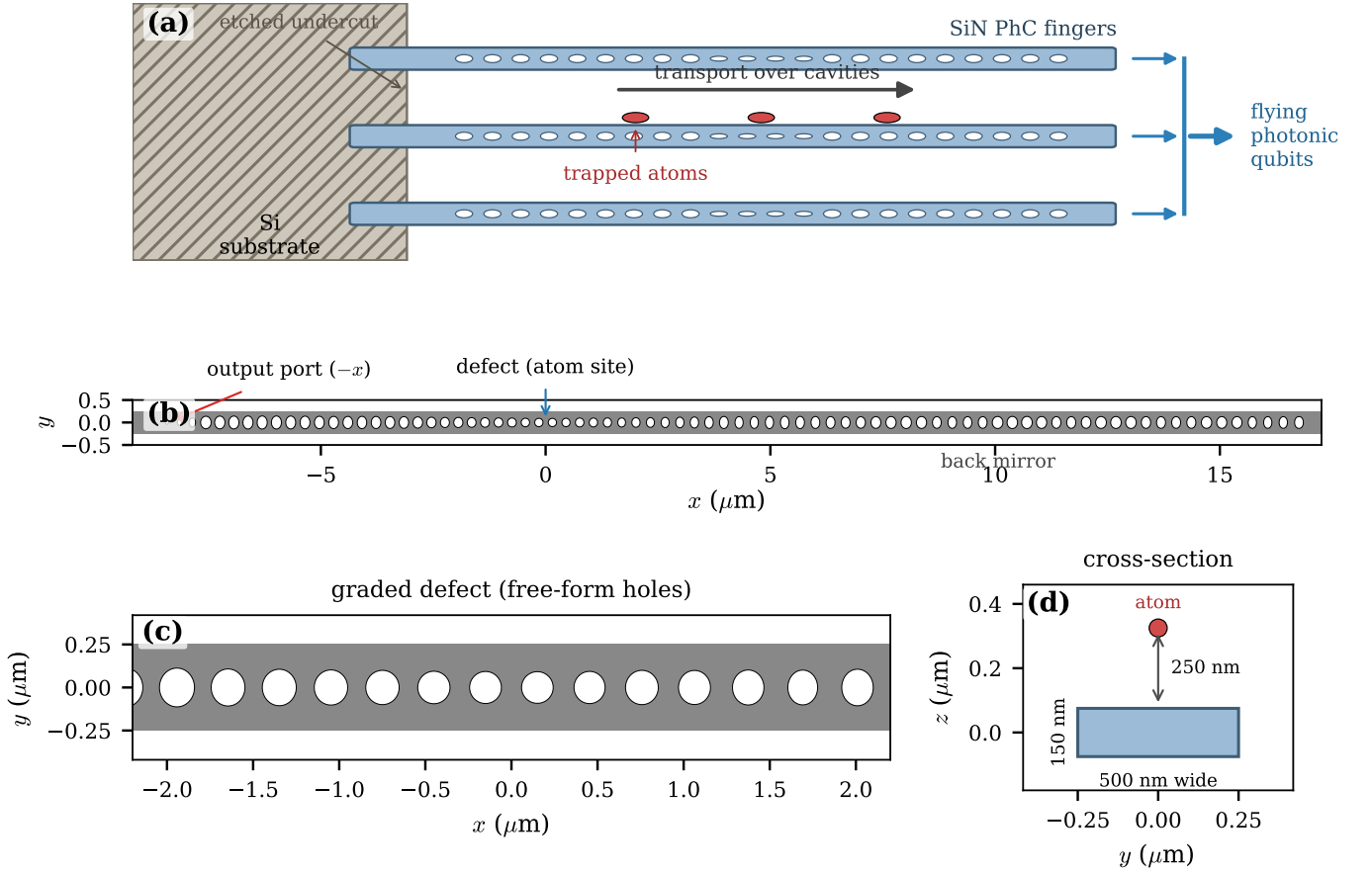


FIG. 1. **Platform and final device.** (a) Concept: an array of suspended SiN photonic-crystal “fingers” cantilevered beyond an etched substrate edge; tweezer-trapped atoms are transported over the series of cavities and transduced into flying photonic qubits collected into fiber. (b) Top view of the inverse-designed cavity, with the asymmetric output ($-x$) coupling mirror, the central defect (atom site), and the long back mirror. (c) Zoom of the graded defect region showing the free-form spline holes and the gentle lattice taper. (d) Beam cross-section: the 500 nm-wide, 150 nm-thick nanobeam with the atom trapped 250 nm above the top surface.

the few-GHz numerical precision of the production mesh, with loaded quality factor $Q = 1.20 \times 10^4$ ($\kappa/2\pi = 32$ GHz), waveguide branching ratio $\beta = 0.886$, cooperativity $C = 24.4$, single-photon coupling $g/2\pi = 1.09$ GHz at the 250 nm trap height, mode volume $V = 0.87 (\lambda/n)^3$, and Purcell factor $F_P \approx 1.0 \times 10^3$. The resulting interface efficiency is $\eta = 0.842$. The loss budget is dominated by the intended channel: 88.6% of the cavity decay exits the output waveguide ($\beta = 0.886$), with only 11% residual scattering, confirming the device is strongly over-coupled rather than merely high- Q .

Figure 4 places this result in context. Panels (a)–(d) show the verified metrics climbing from the seed: the optimizer raises η and β by an order of magnitude past the hand-tuned baseline (dashed), while the cooperativity settles near $C \sim 25$ and the bare Q is deliberately *reduced* from the seed’s value—the design trades useless intrinsic Q for a useful, over-coupled interface. The design lineage [Fig. 4(e),(f)] makes the same point across four geometries: the hand-tuned baseline and the band-engineered seed are both strongly under-coupled

($\beta \sim 0.09$, $\eta \sim 0.08$), even though the seed achieves a nominally enormous intrinsic quality factor; gradient inverse design opens the output port to $\beta = 0.88$ and $\eta = 0.83$, and the conformal trim then lands the resonance on the atomic line [Fig. 4(g)] without disturbing that interface. Panel (h) shows the over-coupling trade-off directly: as the output mirror is weakened the cooperativity rises but β and η fall, so the efficiency optimum [Eq. (6)] sits at intermediate coupling, not at maximal C . We caution that the band-engineered seed’s apparent $Q \sim 10^7$ and $C \sim 10^5$ are unresolved upper bounds—the field barely decays within the simulation window—and not operating values; the operating design is the resolved- Q , over-coupled trimmed cavity.

Relative to the hand-tuned baseline, the final design improves the transduction efficiency by a factor of ten ($\eta : 0.083 \rightarrow 0.842$) and clears both the $\beta \geq 0.80$ and $\eta \geq 0.75$ targets. It is single-mode—the nearest competing resonance lies more than 1 THz away—and the bad-cavity hierarchy $\kappa \gg g \gg \gamma$ is maintained. Across a deterministic fabrication corner grid (hole-radius etch

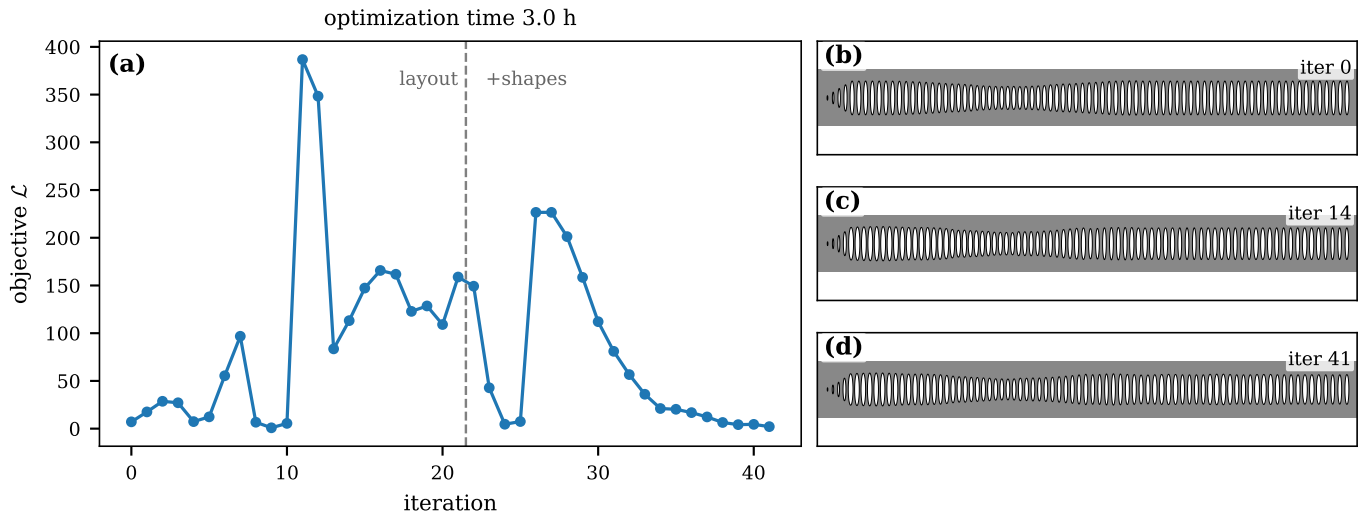


FIG. 2. **Inverse-design trajectory.** (a) Objective $\mathcal{L}$ [Eq. (7)] versus iteration; the dashed line marks the layout $\rightarrow$ shapes stage boundary. The mid-run excursion is the frequency penalty acting as the resonance temporarily drifts off target. (b)–(d) Top-view snapshots of the structure at early, intermediate, and final iterations, showing the evolving hole schedule.

bias ± 4 nm, lattice constant ± 2 nm, beam width ± 10 nm) the efficiency stays in the range $\eta = 0.82$ – 0.86 and the branching ratio $\beta = 0.86$ – 0.89 : the perturbations shift the absolute resonance, which the same global-scale trim restores, but leave the interface metrics essentially intact.

E. Likely operational impact

Interpreted as electromagnetic proxies, these numbers indicate a substantial improvement for the atom-to-flying-qubit channel. The order-of-magnitude gain in η , from ~ 0.07 in single-atom nanocavity demonstrations [12] to 0.84 here, would—taken at face value in a two-node heralding protocol whose rate scales as η^2 —correspond to more than two orders of magnitude in entanglement rate. The cooperativity $C \approx 24$ places the residual cavity-protocol infidelity $\sim \ln C/C \approx 13\%$ in the regime of leading demonstrations, while the strong over-coupling ensures that the emitted photon is delivered to a single, well-defined waveguide mode suitable for fiber collection and interference with a remote node. These statements are projections from the classical interface metrics, not results of a quantum-dynamics simulation.

IV. DISCUSSION

We have inverse-designed a suspended-SiN photonic-crystal nanobeam cavity, resonant with the ^{87}Rb D2 line, that directly optimizes the atom-to-waveguide transduction efficiency. The approach factorizes cleanly: Bloch band engineering supplies a clean photonic foundation with the mirror gap on the atomic line, while adjoint in-

verse design supplies the over-coupled, on-resonance interface; a conformal in-plane scale lands the mode on the line without disturbing the dimensionless design. The result, $\eta = 0.84$ with $\beta = 0.89$ and $C = 24$, is a ten-fold efficiency gain over a hand-tuned baseline and an order of magnitude above prior single-atom nanocavity efficiencies.

Several limitations bound these claims. First and most importantly, g , C , β , and η are scalar proxies extracted from classical electromagnetic fields; they are not outputs of a master-equation or quantum-trajectory cavity-QED simulation, and we make no claim about gate, readout, or entanglement fidelities, atom motion, or multilevel dynamics. Second, the *absolute* resonance frequency is mesh-limited at the tens-of-GHz level, so we report the design at a single fixed production mesh and rely on the demonstrated in-plane trim to absorb the residual; the interface metrics themselves are mesh-robust. Third, the band-engineered seed’s high Q and C are unresolved upper bounds rather than operating values. Fourth, the atom-position maps hold κ and β fixed at the cavity values and re-derive only the local field. Finally, the device has not been fabricated, and the trim is in-plane only, fixed by the deposited film thickness.

These limitations point to natural next steps. A full cavity-QED treatment—a master-equation model including the multilevel atom, motion, and realistic decoherence—would convert the present interface proxies into protocol-level fidelity and rate predictions. Fabrication and cold-atom characterization would test the design against real disorder and the trap geometry; mesh-converged or eigenmode-based frequency solves, together with post-fabrication thermal or MEMS tuning, would pin the absolute resonance to within a linewidth. Finally, the single-cavity design is the unit cell of the finger-array architecture of Fig. 1(a): extending the optimization to

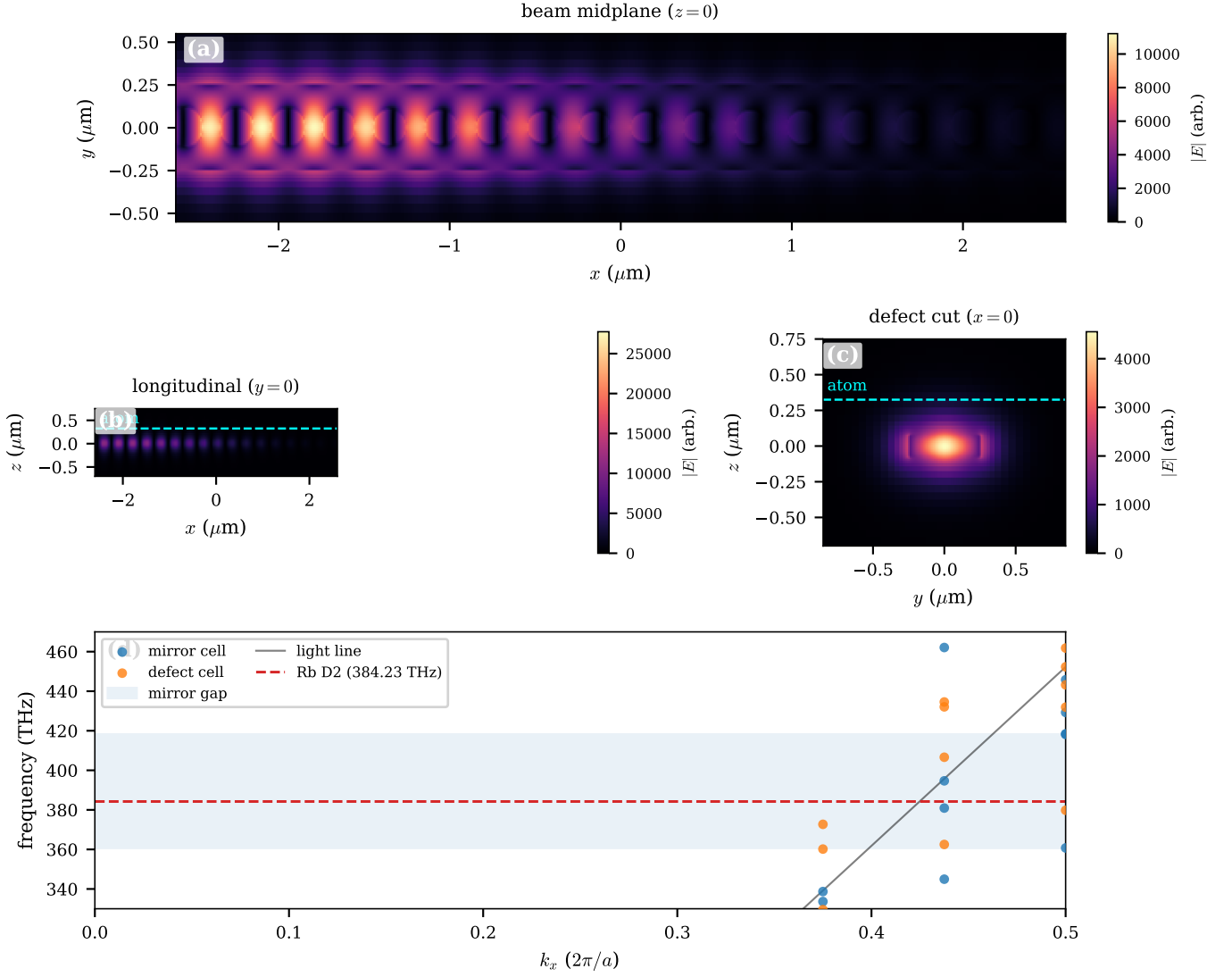


FIG. 3. **Cavity mode and band structure.** (a)–(c) Electric-field magnitude $|E|$ of the final mode in the beam midplane ($z = 0$), the longitudinal plane ($y = 0$), and the defect cross-section ($x = 0$); the dashed line marks the 250 nm atom height, where the evanescent field couples to the trapped atom. The mode is tightly localized at the defect. (d) TE Bloch band structure of the mirror and defect unit cells. The mirror band gap (shading) brackets the Rb D2 line (dashed), confining the mode, while the defect band edge sits just below it; the light line is shown for reference.

multi-finger arrays with shared collection, and allowing a second adjoint stage to push β further, are the path toward a scalable atom-array photonic interconnect.

RESOURCES

Hardware. Electromagnetic simulations were run on the TIDY3D cloud finite-difference time-domain solver (GPU-accelerated, billed in FlexCredits); geometry construction, optimization bookkeeping, and postprocessing ran on a local workstation.

Software. Differentiable FDTD and band-structure

simulations used TIDY3D 2.11.2 with its native automatic-differentiation (adjoint) capability [22]; gradients were propagated with AUTOGRAD and the optimization used Adam. Geometry, postprocessing, and figures used NUMPY, SCIPY, SHAPELY (polygon validity), GDS-FACTORY (GDS export), and MATPLOTLIB. Bloch band structures were computed with TIDY3D unit-cell simulations.

Data and code availability. The inverse-design pipeline, the verified datasets underlying every figure, and a one-command regeneration script are available from the authors on reasonable request.

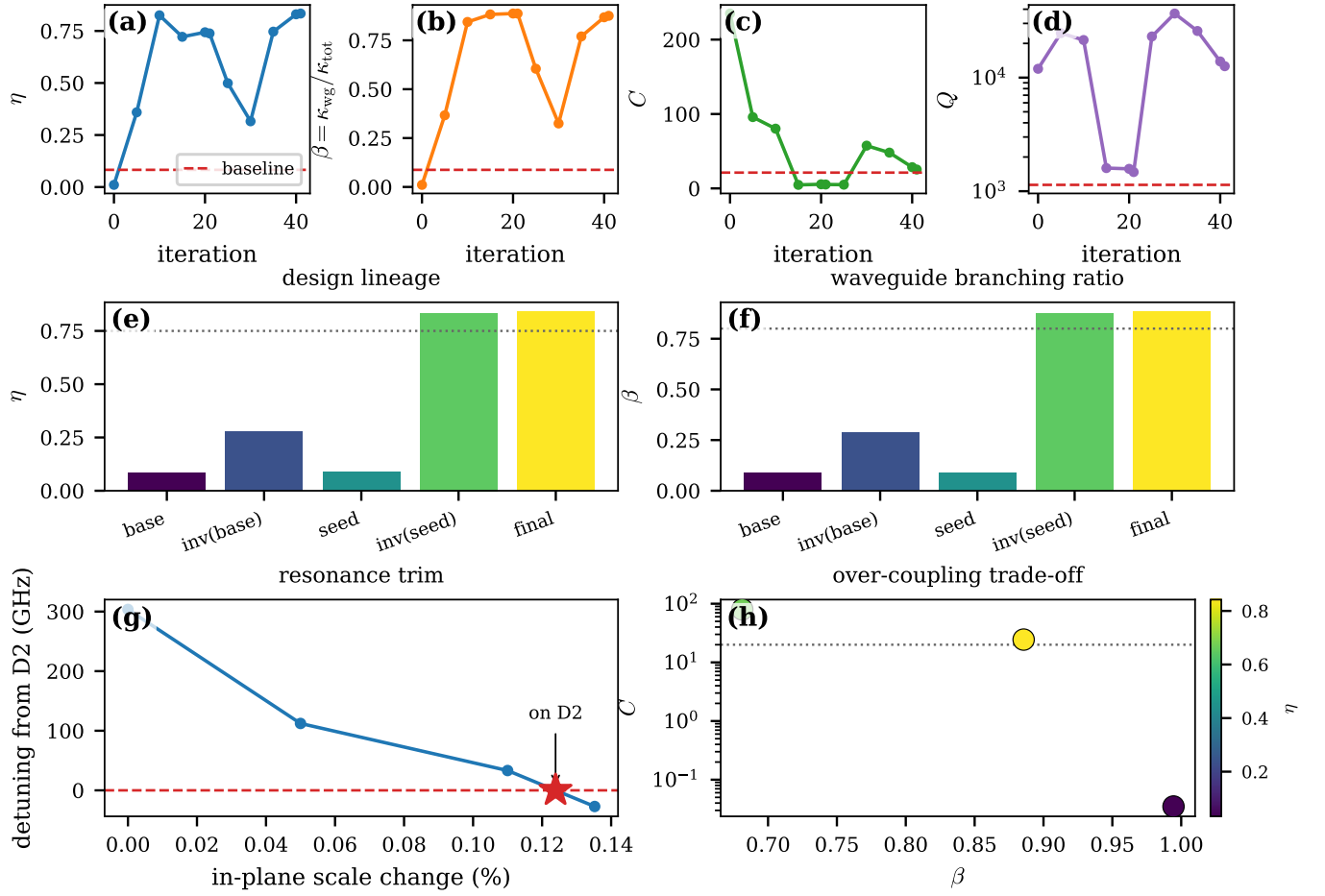


FIG. 4. **Performance evolution, lineage, and tuning.** (a)–(d) Verified η , β , C , and Q versus iteration; dashed lines mark the hand-tuned baseline. The optimizer raises η and β by an order of magnitude while trading the seed’s excess intrinsic Q for a useful, over-coupled interface. (e), (f) Efficiency η and branching ratio β across the design lineage (baseline; gradient-only; band-engineered seed; seeded inverse design; final trimmed); dotted lines mark the 0.75 and 0.80 targets. (g) Resonance trim: detuning from the D2 line versus a global in-plane scale; the star marks the selected on-resonance design. (h) Over-coupling trade-off: cooperativity C versus branching ratio β (color: η) as the output mirror is varied, showing that maximal η does not coincide with maximal C .

- [1] M. Endres, H. Bernien, A. Keesling, H. Levine, E. R. Anschuetz, A. Krajenbrink, C. Senko, V. Vuletić, M. Greiner, and M. D. Lukin, Atom-by-atom assembly of defect-free one-dimensional cold atom arrays, *Science* **354**, 1024 (2016).
- [2] S. Ebadi, T. T. Wang, H. Levine, A. Keesling, G. Semeghini, A. Omran, D. Bluvstein, R. Samajdar, H. Pichler, W. W. Ho, S. Choi, S. Sachdev, M. Greiner, V. Vuletić, and M. D. Lukin, Quantum phases of matter on a 256-atom programmable quantum simulator, *Nature* **595**, 227 (2021).
- [3] S. J. Evered, D. Bluvstein, M. Kalinowski, S. Ebadi, T. Manovitz, H. Zhou, S. H. Li, A. A. Geim, T. T. Wang, N. Maskara, H. Levine, G. Semeghini, M. Greiner, V. Vuletić, and M. D. Lukin, High-fidelity parallel entangling gates on a neutral-atom quantum computer, *Nature* **622**, 268 (2023).
- [4] D. Bluvstein, H. Levine, G. Semeghini, T. T. Wang, S. Ebadi, M. Kalinowski, A. Keesling, N. Maskara, H. Pichler, M. Greiner, V. Vuletić, and M. D. Lukin, A quantum processor based on coherent transport of entangled atom arrays, *Nature* **604**, 451 (2022).
- [5] L.-M. Duan and C. Monroe, Colloquium: Quantum networks with trapped ions, *Rev. Mod. Phys.* **82**, 1209 (2010).
- [6] H. J. Kimble, The quantum internet, *Nature* **453**, 1023 (2008).
- [7] A. Reiserer and G. Rempe, Cavity-based quantum networks with single atoms and optical photons, *Rev. Mod. Phys.* **87**, 1379 (2015).
- [8] A. González-Tudela, A. Reiserer, J. J. García-Ripoll, and F. J. García-Vidal, Light-matter interactions in quantum nanophotonic devices, *Nat. Rev. Phys.* **6**, 166 (2024).
- [9] A. Suleymanzade and I. Dimitrova, Quantum optical

- interfaces for neutral atom arrays, [arXiv:2402.05751](https://arxiv.org/abs/2402.05751) [10.48550/arXiv.2402.05751](https://arxiv.org/abs/2402.05751) (2024).
- [10] S. G. Menon, N. Glachman, M. Pompili, A. Dibos, and H. Bernien, An integrated atom array–nanophotonic chip platform with background-free imaging, *Nat. Commun.* **15**, 6156 (2024).
 - [11] S. G. Menon, K. Singh, J. Borregaard, and H. Bernien, Nanophotonic quantum network node with neutral atoms and an integrated telecom interface, *New J. Phys.* **22**, 073033 (2020).
 - [12] J. D. Thompson, T. G. Tiecke, N. P. de Leon, J. Feist, A. V. Akimov, M. Gullans, A. S. Zibrov, V. Vuletić, and M. D. Lukin, Coupling a single trapped atom to a nanoscale optical cavity, *Science* **340**, 1202 (2013).
 - [13] S. G. Menon, *Combining Nanophotonic Cavity-Arrays with Atomic Arrays: Scaling Quantum Computers and Networks*, Ph.D. thesis, University of Chicago (2025).
 - [14] J. P. Covey, A. Sipahigil, S. Szoke, N. Sinclair, M. Endres, and O. Painter, Telecom-band quantum optics with ytterbium atoms and silicon nanophotonics, *Phys. Rev. Appl.* **11**, 034044 (2019).
 - [15] S. Molesky, Z. Lin, A. Y. Piggott, W. Jin, J. Vucković, and A. W. Rodriguez, Inverse design in nanophotonics, *Nat. Photonics* **12**, 659 (2018).
 - [16] A. Y. Piggott, J. Lu, K. G. Lagoudakis, J. Petykiewicz, T. M. Babinec, and J. Vucković, Inverse design and demonstration of a compact and broadband on-chip wavelength demultiplexer, *Nat. Photonics* **9**, 374 (2015).
 - [17] Q. Quan and M. Lončar, Deterministic design of wavelength scale, ultra-high Q photonic crystal nanobeam cavities, *Opt. Express* **19**, 18529 (2011).
 - [18] Y. Tanaka, T. Asano, and S. Noda, Design of photonic crystal nanocavity with Q-factor of $\sim 10^9$, *J. Lightwave Technol.* **26**, 1532 (2008).
 - [19] A. Bazin, R. Raj, and F. Raineri, Design of silica encapsulated high-Q photonic crystal nanobeam cavity, *J. Lightwave Technol.* **32**, 952 (2014).
 - [20] E. M. Purcell, Spontaneous emission probabilities at radio frequencies, *Phys. Rev.* **69**, 681 (1946).
 - [21] E. N. Knall, C. M. Knaut, R. Bekenstein, D. R. Assumpcao, P. L. Stroganov, W. Gong, Y. Q. Huan, P.-J. Stas, B. Machielse, M. Chalupnik, D. Levonian, A. Suleymanzade, R. Riedinger, H. Park, M. Lončar, M. K. Bhaskar, and M. D. Lukin, Efficient source of shaped single photons based on an integrated diamond nanophotonic system, *Phys. Rev. Lett.* **129**, 053603 (2022).
 - [22] Flexcompute Inc., Tidy3D: Gpu-accelerated electromagnetic solver with native automatic differentiation, <https://www.flexcompute.com/tidy3d/> (2024), version 2.11.2.
 - [23] J. D. Joannopoulos, S. G. Johnson, J. N. Winn, and R. D. Meade, *Photonic Crystals: Molding the Flow of Light*, 2nd ed. (Princeton University Press, 2008).